

Yannik Ammann

Mathematician and software engineer, Zürich
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SUMMARY

Applied mathematician (MSc Mathematics, ETH Zürich) with a software developer's training. I work where statistics and optimisation meet production software and the customers who use it. Today I implement AI invoice automation for customers at BLP Digital AG. Before that I built the stochastic optimisation and risk engine of a portfolio platform at 3rd Eyes Analytics, and Bayesian joint models for oncology at Roche.

EXPERIENCE

Project Manager, AI Automation, BLP Digital AG

July 2026 – present, Zürich

- Own AI invoice-automation projects for several customers, from onboarding and configuration to integration testing and go-live.
- Onboard customers from SMEs to enterprises and program their automation flows in BLP's automation tool.
- Deliver automations in which 80-90% of invoices for delivered products are booked without a human in the loop.
- Run pilots of new automation flows with customers and contribute early-stage work to a self-service product that lets customers set up automations themselves.

Quantitative Developer, 3rd Eyes Analytics

July 2024 – June 2026, Zürich

- Built and maintained a stochastic, risk-aware asset-liability optimisation service for wealth advisory in production.
- Designed core components of the risk engine (Monte-Carlo simulation, scenario analysis) and tuned them for large-scale stochastic simulations in a microservice architecture.
- Led the development of a public API that lets customers embed the quantitative service in their own products. Through the API, the service powered a lead-generation chatbot for financial advice used by about 1,000 users a day.
- Led discovery and feature engineering for risk-management features with wealth-advisory customers, from requirements to delivery.
- Supported pre-sales and onboarding: technical discussions with prospects and customers, turning their requirements into product features.

Data Scientist (Master's thesis project), F. Hoffmann-La Roche Ltd

March 2023 – August 2023, Basel

- Built Bayesian joint models of survival and non-linear longitudinal (mixed-effects) data for oncology, working with limited clinical data.
- Co-developed the open-source Julia package JointSurvivalModels.jl with Roche statisticians and engineers.
- The software enables excluding patients who are unlikely to respond from studies, which lowers patient risk and study cost.
- Worked with clinical scientists, statisticians and the engineering department to turn research requirements into a usable statistical tool.

Teaching Assistant (30%), ETH Zürich

September 2018 – January 2019 and September 2020 – January 2021, Zürich

- Taught classes of 20-25 students in informatics (C++) and geometry, graded coursework and gave feedback.

Full-stack Web Developer, arteria GmbH

August 2014 – July 2015, Reinach BL

- Developed and integrated custom features in web applications, web APIs and CLI tools (Python, Django, JavaScript).

- Maintained, extended and tested internal and open-source packages (django-hijack, virtualenv-mgr).

EDUCATION

Master of Science ETH in Mathematics, ETH Zürich

August 2022 – March 2024

Master's thesis graded 6.0 of 6; overall grade about 5.0 of 6. Thesis: *Joint models of survival and non-linear longitudinal sub-models in a Bayesian framework*, written in collaboration with Roche. Focus on applied mathematics: statistics, linear optimisation and probabilistic algorithms.

Bachelor of Science ETH in Mathematics, ETH Zürich

September 2017 – February 2022

Specialised in computational statistics, machine learning and the mathematics of data science. Bachelor thesis on paraconsistent logic.

Adult Matura (Passerelle), Passerelle Aarau

September 2016 – August 2017

EFZ Software Developer (Applikationsentwickler), Informatik Mittelschule Baden

August 2011 – July 2014

Software development training: Java, object-oriented programming, databases, APIs; foundations in economics, business and Swiss law.

SKILLS

- **Programming:** Python, Julia, SQL, R, C/C++, Java, Bash, JavaScript, HTML, CSS
- **Quantitative methods:** Bayesian inference and probabilistic programming, survival and longitudinal (mixed-effects) models, Monte-Carlo and stochastic simulation, stochastic and linear optimisation, portfolio risk metrics and scenario analysis, numerical integration and ODEs, computational statistics, machine learning, feature engineering
- **Engineering:** REST API design, microservices, Docker, cloud deployment, Git, Linux, testing and CI, Django
- **Delivery:** project management and customer onboarding, requirements engineering and discovery, pre-sales support, AI process automation

LANGUAGES

- **German:** native
- **English:** fluent (C1)

INTERESTS

Landscape and architecture photography, travelling, cooking, philosophy and logic; combat sports, salsa, skiing, ski touring and climbing.